DNA extraction v9.0 (modified BOMB)

Huang MEGA Lab (NCKU)
Lab plain-PDF edition, rebuilt 2026-10-20

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Abstract

DNA extraction using yttria-stabilized zirconia beads lysing and automated magnetic bead-based extraction.

Procedure

SAMPLE COLLECTION

1. (30 s) Add **200 µL** of **0.5 mm beads** to a 2mL screw tube.



2. (30 s) Add **200 µL** of **1 mm beads** to a 2mL screw tube.



3. (30 s) Add **870 µL** Lysis master mix to 2mL screw tube. The final look:



Note In 11F, 4°C fridge

Lysis master mix: **225 µL** of TE buffer + **375 µL** of lysis buffer + **270 µL** of 10M ammonium acetate

4. (1 min) Collect **20-50 mg** of **sample** to 2mL screw tube

Note You can collect up to 100 mg of sample if you can until you bump into the low DNA quality or PCR success rate; by then it means too many inhibitors in the sample and you have to lower the input.

SAMPLE LYSIS

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5. (4 min) Put the 2mL screw tube on vortex for sample lysis, at 3200 rpm **4 min**

Note Remember to balance if you have odd number of samples

CENTRIFUGATION

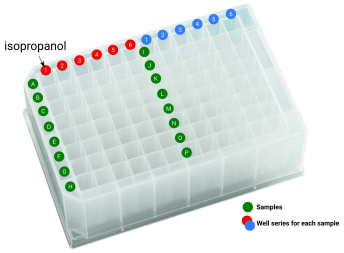
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6. (3 min) Put 2mL screw tube in centrifuge for centrifugation, at this condition: **10 × g, 25 °C, 3 min**

DNA EXTRACTION

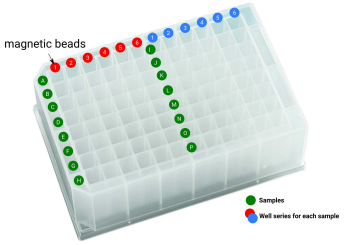
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7. (30 s) Add **350 µL** of **isopropanol** to the 1st well of 96 well plate



DNA PURIFICATION

0.21. (30 s) Add **100 µL** of **magnetic beads** (10 mg/ml) to the 1st well of 96 deep well plate

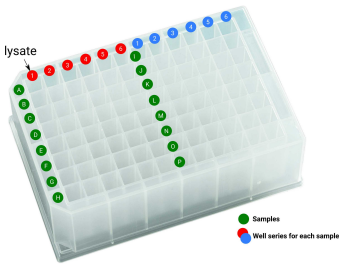


Note Vortex the bottle and pipetting before using magnetic beads; re-do vortex after adding to 3 samples to prevent set down of magnetic beads

DNA EXTRACTION

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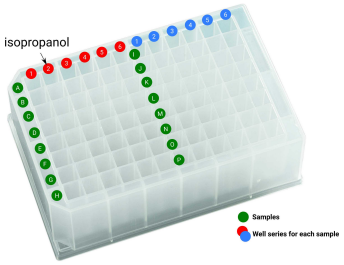
14. (30 s) Add **300-500 µL** of the **sample (lysate)** from the 1.5mL centrifuged tube to the 1st well of 96 deep well plate



Note Pipetting **as many lysate as you can**, as long as it's free of any cell debris (no solids in your tip)

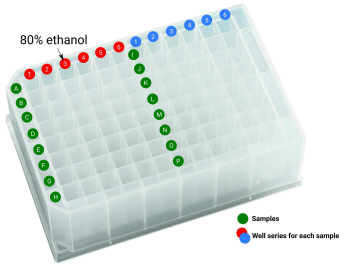
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9. (30 s) Add **400 µL** of **isopropanol** to the 2nd well of 96 deep well plate



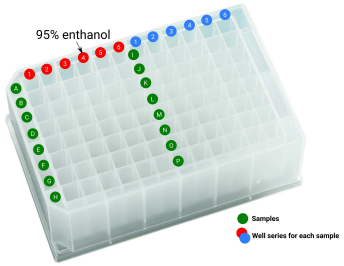
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10. (30 s) Add **300 µL** of **80% ethanol** to the 3rd well of 96 deep well plate



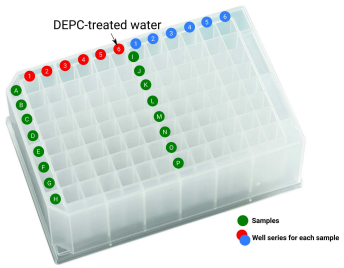
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11. (30 s) Add **300 µL** of **95% ethanol** to the 4th well of 96 deep well plate



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13. (30 s) Add **100 µL** of **DEPC-treated water** to the 6th well of 96 deep well plate



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15. (34 min) Put the prepared 96-deep well plate in the automated DNA extraction machine (ZiXpress 32) and

set up the settings as below.

GDNA COLLECTION

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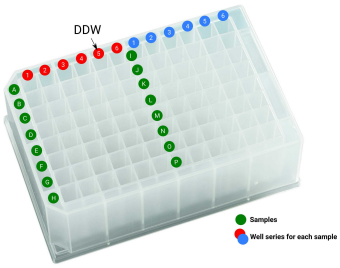
17. After the extraction is done, put on the 96 magnetic plate to pellet the magnetic bead residues.



DNA EXTRACTION

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12. Add **300 µL** of **DDW** to the 5th well of 96 deep well plate



GDNA COLLECTION

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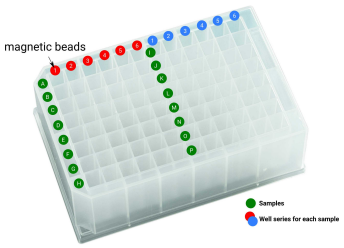
18. Collect **100 µL** of the **eluted sample** (avoid getting magnetic bead) as the DNA template for downstream experiments



DNA EXTRACTION

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8. Add **50 µL** of **magnetic beads (10mg/ml)** to the 1st well of 96 deep well plate



PROGRAM SETTINGS

16. The automated extraction program settings for ZiXpress 32.

1	**0**	**10**	**1000**	**3**	**120**	**0**
2	0	2	400	3	60	0
3	0	2	300	3	60	0
4	15	0	0	0	0	0
4	0	2	300	3	60	0
5	0	1	300	3	0	0
6	0	5	100	3	120	55

Materials

#	**Material**
1	Lysis master mix 870 ml (870 µl/sample)
2	TE buffer stocks 225 ml
3	Lysis buffer stocks 375 ml
4	**10M Ammonium acetate 270 ml**

TE buffer	**225 µl**
Lysis buffer	**375 µl**
10M Ammonium acetate	**270 µl**

1M Tris-HCl pH8.0	**2.25 ml**	**10mM**
1M EDTA	**0.225 ml**	1mM
ddH2O	adjust the volume with water to 225 ml	

Note Prepare 1M Tris-HCl and 1M EDTA first.

Tris-HCl	**7.88 g**	**50 ml**
EDTA	14.61	**50 ml**

Note adjust the pH of EDTA with solid NaOH or 1M NaOH to 8.0. The undissolved EDTA will be dissolved entirely when the pH reaches 8.0.

Guanidine thiocyanate (GITC)	**177.3 g**	**4M**
1M Tris HCl pH8.0	**18.75 ml**	50mM
Sodium Lauryl Sulfate	3.75 g	0.5 g
1M EDTA	**7.5 ml**	20mM
ddH2O	adjust the pH with HCl to 7.6–8.0 and the volume with water to 375 ml	

Ammonium acetate	**208.116 g**
ddH2O	adjust the volume with water to 270 ml

Note Ammonium acetate is hygroscopic. Do not add more than half of the water needed to dissolve it.